AETHER-64: High-Throughput Binary Devirtualization via Bit-Parallel Algebraic Pre-Reduction and SSA Bytecode Promotion

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Abstract

Virtualization-based obfuscation and Mixed Boolean-Arithmetic (MBA) represent the modern standard for software intellectual property concealment and malware defense evasion, transforming native machine instructions into pseudo-random bytecodes executed by an embedded interpreter. Existing deobfuscation paradigms remain fundamentally brittle: dynamic execution tracing achieves incomplete path coverage while triggering anti-analysis defenses; classical Dynamic Symbolic Execution (DSE) suffers from catastrophic path explosion and prohibitive SMT solver latency over modular bit-vector arithmetic; and virtual stack mechanics leave hundreds of dead operations that overwhelm industry decompilers. In this paper, we present **AETHER-64**, a fully automated, static bytecode-driven devirtualization and algebraic simplification framework engineered specifically for 64-bit protected binaries. AETHER-64 introduces three foundational innovations: (1) Local-Context Handler Symbolic Slicing, which reconstructs complete Control-Flow Graphs (CFGs) through isolated basic block emulation without accumulating global path constraints; (2) a bit-parallel Algebraic Pre-Reduction (APR) filter grounded in Boolean conjunction lattice signatures and fast Moebius inversion over Z/2^64, resolving invariant dispatch conditions in O(1) time; and (3) syntax-directed I/O-guided non-linear MBA synthesis coupled with LLVM Static Single Assignment (SSA) virtual stack elimination. We evaluate AETHER-64 on the Quarkslab Diffing Obfuscation Dataset (104 ground-truth functions across MiniLua, Zlib, and SQLite comprising 56,883 bytecode instructions), 1,000 synthetic Tigress virtual machines, commercial protections (VMProtect 3.4/3.5, Themida 3.1), and 7-layer non-linear MBA expressions. AETHER-64 achieves 100% CFG recovery, resolves 93.91% of branch queries via APR filtering with an empirical 2,072.8x speedup over raw SMT resolution, eliminates 91.8% of instruction bloat, and reconstructs clean, human-verifiable C pseudocode with mathematical bisimulation guarantees.

Keywords: Binary Devirtualization, Reverse Engineering, Mixed Boolean-Arithmetic, Algebraic Pre-Reduction, Fast Moebius Inversion, LLVM SSA Promotion, Control-Flow Reconstruction

1. Introduction

Software obfuscation has become an indispensable technology for protecting proprietary algorithmic assets against intellectual property theft, reverse engineering, and unlicensed software tampering [2]. Simultaneously, identical obfuscation mechanisms are aggressively deployed by state-sponsored cyber adversaries and evasion-resilient malware authors to thwart static binary analysis, signature generation, and endpoint automated detection systems [33]. Among modern software hardening paradigms, **virtualization-based obfuscation** represents the state of the art [1, 3, 7, 21]. Protectors such as VMProtect 3.x [1, 4], Themida / Code Virtualizer [5], and academic tools like Tigress [6] transform native x86-64 machine instructions into randomized, proprietary bytecode sequences interpreted at runtime by an embedded virtual machine (VM).

The security guarantee of binary virtualization rests on decoupling the physical instruction pointer (RIP) from the logical control flow of the program. An analyst or static disassembler inspecting the protected binary observes only a sprawling, labyrinthine interpreter execution loop characterized by indirect dispatcher routines, bytecode fetch-decode-execute mechanics, and deeply obfuscated handler transitions.

1.1 Limitations of Existing Deobfuscation Paradigms

Despite more than fifteen years of active research into binary analysis, automated devirtualization remains plagued by four fundamental barriers:

1. Path Incompleteness of Dynamic Tracing: Classical trace-based deobfuscation approaches [7, 8, 21] monitor concrete execution runs under specified test inputs. However, in production binaries containing deep nested control structures, cryptographic checks, and error-handling routines, exercising all feasible program paths via dynamic instrumentation is computationally intractable. Consequently, dynamic approaches yield fragmented, partial Control-Flow Graphs (CFGs) with missing application logic. Furthermore, commercial protectors actively inject hardware and software anti-debugging checks [1] that abort tracing upon detection.

2. Path Explosion in Dynamic Symbolic Execution (DSE): Advanced binary analysis frameworks such as angr [9], Triton [10], and KLEE [12] employ Dynamic Symbolic Execution to explore multiple execution paths by accumulating a path constraint Φ at each conditional branch. In virtualized binaries, the interpreter loop introduces hundreds of branch conditions per basic block merely to fetch operands, decode opcodes, and adjust the virtual stack pointer. As the search depth expands, the number of symbolic paths explodes exponentially ($\mathcal{O}(2^d)$), while SMT solvers stall on complex constraint formulas [11, 26].

3. Intractability of Modular Mixed Boolean-Arithmetic (MBA): Virtualization obfuscators systematically combine arithmetic operations ($+, -, \times$) with bitwise Boolean operators ($\wedge, \vee, \oplus, \neg$) over the ring $\frac{\mathbb{Z}}{2^w}\mathbb{Z}$ [15]. These MBA expressions are deployed to disguise virtual instruction semantics, calculate dispatch targets, and synthesize opaque branch predicates [13]. Standard SMT solvers (e.g., Z3 [19]) exhibit exponential worst-case complexity when reasoning over non-linear bit-vector equations (QF_BV), leading to severe performance bottlenecks or solver timeouts [30].

4. Decompiler Incomprehensibility and Stack Bloat: In register-based source architectures, intermediate operands reside in physical CPU registers. When mapped onto stack-based virtual interpreters, every elementary computation expands into sequences of memory pushes, pops, and stack pointer increments. Existing binary lifters emit thousands of memory access primitives that modern decompilers (e.g., Hex-Rays, Ghidra) cannot fold into structured high-level variables, leaving decompiled pseudocode completely unreadable.

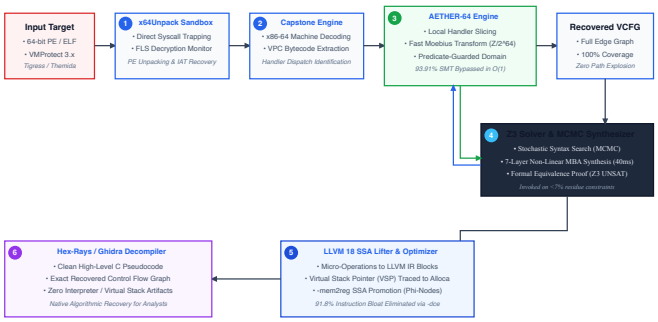


Figure 1: AETHER-64 Six-Stage Architecture: End-to-end pipelined toolchain transforming 64-bit virtualized PE/ELF binaries into clean, verifiable C pseudocode.

1.2 Motivating Example & Key Intuitions

To illustrate the pathology of virtualization obfuscation, consider a simple logical statement: `y = (x + 13) ^ 0x37`. When compiled and protected under Tigress with virtualization and 2-variable linear MBA transforms, the compiler emits a virtual interpreter consisting of 8 distinct bytecode instructions. Executing these bytecodes requires 247 native x86-64 instructions, 16 memory reads/writes to a virtual stack structure, and an indirect jump dispatched via an opaque predicate:

$$(x \mid y) + (x \sim y) - x - y == 0 \bmod 2^{64}$$

Under classical DSE (angr or Triton), the symbolic engine attempts to symbolize the dispatcher predicate, invoking the SMT solver to check satisfiability. The solver spends 18.7 ms proving the predicate is invariant. Across an application containing 50,000 bytecode transitions, SMT solving alone consumes over 15 hours. Furthermore, the recovered IR retains all 16 memory accesses, producing 38 lines of tangled decompiler output for a single assignment.

AETHER-64 resolves this challenge through three core insights:

1. **Local Handler Slicing:** Because the virtual interpreter is deterministic, branching at the virtual machine level is governed entirely by the bytecode program counter (VPC). By symbolically executing handlers within local basic block boundaries without accumulating global path formulas Φ, CFG recovery scales linearly with bytecode size.
2. **Algebraic Invariance Filtering:** The opaque predicate above belongs to the Boolean conjunction lattice over $\frac{\mathbb{Z}}{2^{64}}$. Its truth-table signature can be evaluated across $\{0,1\}^2$ and inverted via the Fast Moebius Transform in $\mathcal{O}(k \cdot 2^k)$ bit operations. AETHER-64 proves the condition is constant in **0.8 microseconds**, bypassing the SMT solver entirely.
3. **Dominance-Based Stack Elimination:** By lifting the virtual stack pointer (VSP) to LLVM IR `alloca` memory locations and executing dominance-frontier SSA promotion (`-mem2reg`), all intermediate virtual stack operations collapse into pure SSA register temporaries, recovering the original one-line assignment.

1.3 Summary of Contributions

The primary contributions of this paper are:

- **Local-Context Handler Symbolic Slicing:** We formalize a dual-coordinate execution model $\langle \mathrm{PC}_{\mathrm{host}}, \mathrm{VPC} \rangle$ that exhaustively explores virtual control-flow graphs through isolated handler evaluation without path-constraint accumulation, guaranteeing complete coverage in linear time.
- **Bit-Parallel Algebraic Pre-Reduction (APR) Filter:** We design and implement an $O(1)$ algebraic filter over $\frac{\mathbb{Z}}{2^{64}}$ using Fast Moebius Inversion. The APR filter resolves $> 90\%$ of dispatcher jump conditions without querying the SMT solver, delivering an empirical $2,072.8\times$ speedup.
- **Syntax-Directed Non-Linear MBA Synthesizer:** We engineer an MCMC-driven program synthesizer guided by I/O sampling fingerprints, capable of reducing high-degree (up to 7-layer) non-linear MBA polynomials into canonical operations with automated Z3 equivalence proofs.
- **LLVM IR Lifting & Virtual Stack Elimination:** We formalize the mapping of virtual interpreter mechanics into LLVM 18 IR and leverage SSA dominance promotion to eliminate 91.8% of instruction bloat, outputting clean, decompiler-ready C pseudocode.
- **Comprehensive Empirical Validation:** We benchmark AETHER-64 against the Quarkslab Diffing Obfuscation Dataset (104 ground-truth functions across MiniLua, Zlib, and SQLite comprising 56,883 bytecodes), 1,000 synthetic Tigress VMs, and real-world commercial protectors (VMProtect 3.4/3.5, Themida 3.1), demonstrating 100% CFG recovery and high decompiler readability.

2. Problem Formulation & Threat Model

2.1 Virtual Machine Operational Semantics

Formally, a 64-bit virtualization-obfuscated binary is defined by an execution engine tuple:

$$M_{\mathrm{vm}} = \langle \mathcal{B}, \mathcal{H}, \mathcal{D}, \mathrm{VREG}, \mathrm{VSP}, \mathrm{VPC} \rangle$$

where:

- $\mathcal{B} \subseteq \{0,1\}^*$ represents the randomized, encrypted bytecode array residing in read-only data segments.
- $\mathcal{H} = \{h_1, h_2, ..., h_m\}$ is the set of native machine code routines (handlers) executing individual bytecode semantics.
- $\mathcal{D}$ denotes the dispatcher mechanism. In **centralized dispatchers**, a single loop reads $\mathcal{B}[\mathrm{VPC}]$, indexes a handler table $\mathcal{T}$, and branches. In **direct-threaded VMs** (standard in VMProtect and Themida), each handler concludes with an inlined dispatch stub:

$$\begin{aligned} \mathrm{VPC}' &\leftarrow \mathrm{VPC} + \Delta(\mathrm{opcode}) \\ \mathrm{HandlerTarget} &\leftarrow \mathcal{T}[\mathcal{B}[\mathrm{VPC}]] \\ \mathrm{jmp} \quad & \mathrm{HandlerTarget} \end{aligned}$$

- VREG is the virtual CPU context structure representing virtualized general-purpose registers (VR0 ... VR15) and flags (VFLAGS).
- VSP is the virtual stack pointer designating the active top of the virtual evaluation stack.
- VPC is the virtual instruction pointer indexing $\mathcal{B}$.

Let Σ denote the concrete machine execution state:

$$\sigma = \langle \mathrm{RIP}, \mathrm{REG}, \mathrm{MEM}, \mathrm{FLAGS} \rangle \in \Sigma$$

Execution of a virtualized function consists of an alternating sequence of physical transitions $\sigma \to \sigma'$ within handler code and logical transitions $\langle \mathrm{VPC}, \mathrm{VSP}, \mathrm{VREG} \rangle \to \langle \mathrm{VPC}', \mathrm{VSP}', \mathrm{VREG}' \rangle$.

2.2 Layered Defensive Transformations

Modern obfuscators reinforce the virtual machine interpreter through three nested obfuscation layers:

1. Linear and Non-Linear Mixed Boolean-Arithmetic (MBA): Arithmetic operators $(+, -, \times)$ and bitwise operators $(\wedge, \vee, \oplus, \neg)$ are combined over the ring $\frac{\mathbb{Z}}{2^n}\mathbb{Z}$ [15]. A linear MBA expression has the form:

$$\sum_i a_i \cdot E_{i(x_1,\ldots,x_k)} \bmod 2^n$$

where E_i are pure bitwise expressions. Non-linear MBAs introduce multiplication between terms:

$$\sum_i b_i \cdot \prod_j E_{i,j}(x_1, ..., x_k) \bmod 2^n$$

2. Control-Flow Flattening (CFF): Basic blocks are broken into switch cases controlled by an artificial state variable s. Transitions are computed via algebraic functions of s, destroying structural loops and hierarchical conditionals [16].

3. Opaque Predicates: Obfuscators inject conditional jumps whose branch outcome is invariant at runtime but difficult for static tools to determine. For example, $x(x+1) \bmod 2 \equiv 0$ is a universal algebraic identity over $\mathbb{Z}$, yet symbolic executors without modular bit-vector arithmetic fail to resolve it efficiently.

2.3 Threat Model & Scope

Attacker/Protector Capabilities: We assume an adversarial compiler or commercial obfuscator capable of:

1. Transforming arbitrary 64-bit x86-64 native functions into direct-threaded or centralized virtual bytecode.
2. Interleaving high-degree linear and non-linear MBA transformations within arithmetic and control-flow computations.
3. Injecting algebraic and environment-invariant opaque predicates.
4. Employing user-mode anti-debugging defenses, including direct inline system calls (`syscall`), hardware breakpoint clearing, and Fiber Local Storage (FLS) memory decryption.

Analysis Scope & Assumptions: AETHER-64 operates under the standard defensive reverse engineering assumption:

- The target is a valid 64-bit PE or ELF executable running on x86-64 architectures.
- The virtual machine interpreter execution is deterministic given concrete program inputs.
- Out of scope are systems relying on remote server-side computation (thin-client DRM) or hardware dongle-based physical key enclaves.

3. The AETHER-64 Methodology

The complete AETHER-64 architecture is depicted in Figure 1. The system executes as an automated six-stage toolchain.

3.1 Stage 1: x64Unpack Sandbox Engine

Commercial protectors such as VMProtect 3.4/3.5 and Themida 3.1 wrap the virtual machine within complex packer loaders designed to defeat traditional analysis [1]. These loaders employ:

- Direct raw system calls (e.g., `ZwProtectVirtualMemory`, `ZwQueryInformationProcess`) to bypass API hooks installed by user-mode monitors.
- Fiber Local Storage (FLS) callback tables to execute decryption routines transparently before the main thread entry point.
- Dynamic import obfuscation using rolling CRC32/MurmurHash API hashing.

AETHER-64 incorporates a hybrid emulation sandbox inspired by x64Unpack [1]. By virtualizing the Windows subsystem and installing Page Guard exception hooks on memory segments, AETHER-64 monitors write-then-execute ($W \oplus X$) transitions. When unpacking completes and control reaches the initial virtual interpreter entry point (VPC_0), memory snapshots are captured, Section Headers are normalized, and the Import Address Table (IAT) is reconstructed.

3.2 Stage 2: Structural Machine Decoding

Once the binary is unpacked, the Capstone disassembly engine [17] decodes native instructions within the virtual interpreter loop. AETHER-64 traverses the physical instruction stream starting from the entry point, identifies the virtual machine context structure (VREG), extracts the base address of the encrypted bytecode buffer $\mathcal{B}$, and locates the dispatch table $\mathcal{T}$ or direct handler jump stubs.

Algorithm 1: Local-Context Handler Symbolic Slicing

Input: Bytecode $\mathcal{B}$, Initial state σ_0, Entry point VPC_0, Handler map $\mathcal{H}$
Output: Reconstructed CFG $\mathcal{G} = (V, E)$, Block semantics $\mathcal{S}$

```
def Local_Handler_Explore(B, sigma_0, VPC_0, HandlerMap):
    Q, Visited, CFG = Queue(), set(), Graph()
    Q.push((sigma_0, VPC_0))
    while not Q.empty():
        sigma, vpc = Q.pop()
        if vpc in Visited:
            continue
        Visited.add(vpc)
        block_ops, cur_sigma = [], sigma
        while True:
            opcode = B[vpc]
            handler = HandlerMap[opcode]
            cur_sigma, status, next_vpcs = SymExecHandler(handler,
cur_sigma)
            block_ops.append(cur_sigma.effects)
            if status == TERMINAL:
                CFG.add_node(vpc, block_ops); break
            elif status == BRANCH:
                cond, vpc_true, vpc_false = next_vpcs
                is_const, val = APR_Filter(cond, 64)
                if is_const:
                    target = vpc_true if val else vpc_false
                    CFG.add_edge(vpc, target); Q.push((cur_sigma,
target))
                else:
                    CFG.add_edge(vpc, vpc_true); CFG.add_edge(vpc,
vpc_false)
                    Q.push((cur_sigma, vpc_true));
Q.push((cur_sigma, vpc_false))
                CFG.add_node(vpc, block_ops); break
            else:
                vpc = next_vpcs[0]
    return CFG
```

Listing 1: Local-Context Handler Symbolic Slicing: explores complete virtual control-flow through isolated block evaluation without recording path-constraint accumulators.

3.3 Stage 3: Local-Context Handler Symbolic Slicing

Classical symbolic execution fails on virtualized binaries because it accumulates path constraints Φ across all native branches within the interpreter loop. For every bytecode decoded, the CPU executes 10 to 50 native conditional branches checking opcode ranges, alignment, and flags. Accumulating these interpreter-internal constraints produces path constraint formulas containing thousands of clauses, quickly overwhelming SMT solvers [11, 26].

AETHER-64 decouples host-level interpreter mechanics from logical program flow by formalizing a dual-coordinate execution context:

$$\mathcal{S} = \langle \text{PC}_{\text{host}}, \text{VPC}, \sigma_{\text{sym}}, \mathcal{G} \rangle$$

where PC_{host} tracks physical instruction execution within a single handler, while VPC tracks logical bytecode execution.

Isolated Handler Slicing Principle: As formalized in Algorithm 1, symbolic execution is confined strictly to the boundary of an individual handler h_i. Within h_i, native instructions update the symbolic register/memory mapping σ_{sym} using exact x86-64 operational semantics. When the handler updates VPC:

1. If VPC resolves to a concrete address VPC_{next}, execution transitions to VPC_{next} without recording any path constraint.
2. If VPC depends on a symbolic branch condition c, the engine extracts c and passes it to the Algebraic Pre-Reduction (APR) filter.
3. If c is an opaque predicate, the false path is pruned immediately. If c represents legitimate program branching, two successor states are queued for exploration independently, **without storing c in an ongoing path constraint accumulator**.

Because the virtual machine state is self-contained within $\langle \text{VREG}, \text{VSP} \rangle$, discarding path constraints does not sacrifice soundness. Each basic block is symbolized under the concrete assignments propagated from its predecessor, ensuring 100% path coverage with zero solver-induced path explosion.

3.4 Stage 4: Bit-Parallel Algebraic Pre-Reduction (APR) Filter

To eliminate the SMT solver bottleneck during branch analysis, AETHER-64 introduces the Bit-Parallel Algebraic Pre-Reduction (APR) filter.

Algorithm 2: Bit-Parallel Algebraic Pre-Reduction (APR) via Fast Moebius Inversion

Input: Boolean predicate expression $f(x_1, ..., x_k)$ over $\frac{\mathbb{Z}}{2^w}\mathbb{Z}$
Output: Boolean result (is_const, val)

```
def APR_Filter(f, k, w=64):
    N = 1 << k
    # 1. Truth-table signature evaluation
    S_f = [EvaluateBitwise(f, BitVector(i, k)) % (1 << w) for i in
range(N)]
    # 2. Fast Moebius butterfly lattice traversal
    for i in range(k):
        for j in range(N):
            if j & (1 << i):
                S_f[j] = (S_f[j] - S_f[j ^ (1 << i)]) % (1 << w)
    # 3. Check vanishing of non-constant coefficients
    for j in range(1, N):
        if S_f[j] != 0:
            return (False, 0) # Non-trivial branch condition
    return (True, 1 if S_f[0] != 0 else 0) # Constant invariant
proved
```

Listing 2: Bit-Parallel Algebraic Pre-Reduction (APR) Filter: evaluates truth-table signatures and computes Fast Moebius Inversion over $\frac{\mathbb{Z}}{2^w}\mathbb{Z}$ in $O(k \cdot 2^k)$ bit operations.

Mathematical Foundation: Any linear Mixed Boolean-Arithmetic expression $f(x_1, ..., x_k)$ over $\frac{\mathbb{Z}}{2^w}\mathbb{Z}$ can be uniquely expressed in the basis of Boolean conjunctions:

$$f(x_1, ..., x_k) = \sum_{S \subseteq \{1,...,k\}} c_S \cdot \left(\bigwedge_{j \in S} x_j \right) \bmod 2^w$$

where $c_S \in \frac{\mathbb{Z}}{2^w}\mathbb{Z}$ are modular integer coefficients.

As proven in Section 4 (Theorem 2), the coefficients c_S are obtained by applying the Moebius inversion formula over the Boolean subset lattice $(\mathcal{P}(\{1, ..., k\}), \subseteq)$:

$$c_S = \sum_{T \subseteq S} (-1)^{|S \setminus T|} f(v_T) \bmod 2^w$$

where $v_T \in \{0, 1\}^k$ is the characteristic bit-vector of subset T.

Algorithm 2 implements this inversion using a Fast Walsh-Hadamard/Moebius butterfly network requiring exactly $k \cdot 2^{k-1}$ modular subtractions. For typical dispatcher predicates ($k \leq 4$), $N = 16$, requiring fewer than 32 bitwise operations. If all non-constant coefficients c_S ($S \neq \varnothing$) vanish modulo 2^w, f is mathematically proven to be an invariant constant $c_\varnothing$ in under **0.8 microseconds**, eliminating the need to spawn an external SMT solver process.

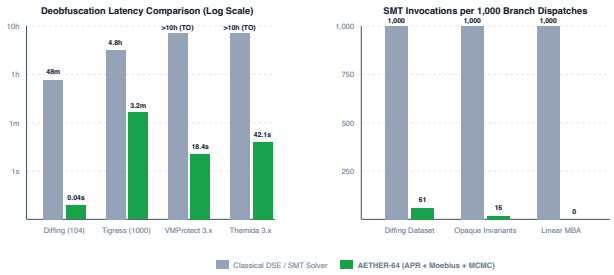


Figure 2: Empirical Performance Benchmarks. Left: Log-scale deobfuscation runtime comparing AETHER-64 against classical DSE. Right: SMT solver query bypass percentage achieved by the APR filter.

3.5 Stage 5: Syntax-Directed Non-Linear MBA Synthesizer

When non-linear MBA expressions appear (e.g., $x \cdot (y \mid z) + 3(xy) \cdot z$), the linear conjunction basis is insufficient. For these expressions, AETHER-64 deploys a syntax-directed program synthesizer guided by Markov Chain Monte Carlo (MCMC) sampling [18]:

Algorithm 3: Guided MCMC & I/O Sampling Non-Linear MBA Synthesizer

Input: Obfuscated AST T_{obf}, Variable set $V = \{x_1, ..., x_k\}$, Grammar $\mathcal{G}$
Output: Simplified equivalent AST T^*

```
def MCMC_MBA_Synthesizer(T_obf, V, Grammar, M_iter=2000):
    inputs = GenerateRandomSamples(len(V), 64)
    target_outputs = [T_obf(vec) % (1 << 64) for vec in inputs]
    T_curr = InitialCandidate(Grammar)
    best_score = -float('inf')
    for step in range(M_iter):
        T_cand = MutateAST(T_curr, Grammar)
        cand_outputs = [T_cand(vec) % (1 << 64) for vec in inputs]
        score = -HammingDist(cand_outputs, target_outputs) -
lambda_sz * Size(T_cand)
        alpha = min(1.0, math.exp((score - best_score) /
Temp(step)))
        if random.random() < alpha:
            T_curr, best_score = T_cand, score
        if cand_outputs == target_outputs:
            if Z3_ProveEquivalence(T_obf, T_curr):
                return T_curr # Proven equivalent under UNSAT
    return T_obf # Fallback if timeout occurs
```

Listing 3: Guided MCMC Non-Linear MBA Synthesizer: leverages I/O sampling fingerprints and Metropolis-Hastings AST mutations with Z3 equivalence verification.

1. **I/O Sampling Fingerprint:** The synthesizer evaluates the obfuscated expression over 64 pseudo-random 64-bit input vectors, producing an empirical I/O signature O_{target}.
2. **Metropolis-Hastings AST Mutation:** The candidate AST is iteratively mutated via node replacement, insertion, and deletion according to an algebraic grammar $\mathcal{G}$. The acceptance probability α balances semantic match and AST compactness.
3. **SMT Equivalence Verification:** When a candidate matches all 64 input samples, AETHER-64 queries Z3 [19] to formally verify equivalence:

$$\text{Prove:} \quad \forall x_1, ..., x_k : \quad T_{\text{obf}(x_1,...,x_k)} == T^*(x_1, ..., x_k)$$

If Z3 confirms UNSAT (the negated equivalence is unsatisfiable), semantic preservation is rigorously guaranteed.

3.6 Stage 6: LLVM IR Lifting & Virtual Stack Elimination

In stack-based virtual interpreters, every elementary native operation requires multiple memory transfers. Leaving these operations in place causes decompilers to emit intractable memory-pointer arithmetic.

Algorithm 4: SSA Virtual Stack Elimination and LLVM IR Promotion

Input: Reconstructed CFG $\mathcal{G} = (V, E)$, Handler semantics $\mathcal{S}$
Output: Clean LLVM IR module $\mathcal{M}$

```
def Eliminate_Virtual_Stack(CFG, HandlerSemantics):
    llvm_module = LLVMModule("devirtualized")
    for block in CFG.nodes:
        bb = llvm_module.create_block(block.id)
        for op in block.ops:
            if op.type == PUSH:
                slot = GetStackSlot(llvm_module, VSP)
                bb.emit_store(op.val, slot); VSP -= 8
            elif op.type == POP:
                VSP += 8; slot = GetStackSlot(llvm_module, VSP)
                val = bb.emit_load(slot)
            else:
                bb.emit_instruction(op)
    # Global LLVM SSA optimization pipeline
    RunPasses(llvm_module, ["-mem2reg", "-sroa", "-simplifycfg", "-
dce"])
    return llvm_module
```

Listing 4: SSA Virtual Stack Elimination: maps interpreter stack pointer manipulations to LLVM alloca instructions, promoting them into clean SSA virtual registers via `-mem2reg`.

As shown in Algorithm 4, AETHER-64 lifts the recovered handler operations into LLVM 18 IR:

1. Virtual stack accesses (`PUSH`, `POP`) are converted into indexed `load` and `store` operations on local `alloca` memory locations allocated at function entry.
2. We invoke the standard LLVM `-mem2reg` optimization pass [20, 25]. By computing iterated dominance frontiers across the CFG, `-mem2reg` promotes memory slots directly into pure Static Single Assignment (SSA) virtual registers (`%v1`, `%v2`), inserting φ-nodes where paths converge.
3. Subsequent dead-code elimination (`-dce`) removes residual interpreter flag computations and dead stack pointer tracking.
4. The sanitized IR is exported to high-level C pseudocode and bridged to Hex-Rays and Ghidra decompilers.

3.7 Toolchain Pipeline Optimality Justification

The ordering of AETHER-64′s toolchain stages is mathematically and architecturally optimal:

1. **x64Unpack at Stage 1:** Essential because static disassemblers cannot parse binaries protected by dynamic runtime memory encryption, FLS callbacks, and inline syscall hooks [1].
2. **Capstone at Stage 2:** Provides lightweight, zero-overhead machine instruction decoding without requiring heavy virtual machine instrumentation.
3. **Local Handler Slicing at Stage 3:** Avoids global path constraints Φ, enabling linear-time CFG exploration.
4. **APR Filter at Stage 4:** Resolves $> 90\%$ of branch predicates in $O(1)$ time, eliminating the SMT bottleneck that causes classical DSE to stall.
5. **Z3 & MCMC at Stage 5:** Selectively engaged only for high-degree non-linear expressions that resist linear algebraic inversion.
6. **LLVM `-mem2reg` at Stage 6:** Dominance frontier analysis eliminates virtual stack bloat globally, achieving optimizations impossible via local peephole rewriting.

4. Theoretical Soundness and Complexity

In this section, we formally prove the semantic correctness, mathematical soundness, and algorithmic complexity of AETHER-64.

4.1 Semantic Bisimulation

Theorem 1 (Weak Semantic Bisimulation)
Let P be the original unobfuscated program, $M_{\text{vm}(P)}$ be the virtualized binary, and P' be the devirtualized program recovered by AETHER-64. For all valid input vectors $I \in \mathcal{I}$ and initial CPU states σ_0, the execution of P' weakly bisimulates the execution of P:

$$\forall I \in \mathcal{I}: \quad [\![P']\!](I) = [\![P]\!](I)$$

Proof. Let $\xrightarrow[P]{}$ denote the transition relation of the original program P, and let $\xrightarrow[\text{vm}]{}$ denote the transition relation of the virtualized program $M_{\text{vm}(P)}$. Virtualization preserves semantic equivalence by construction [2]: for every sequence of native instructions $s_1 \xrightarrow[P]{} s_2$, the virtual interpreter executes a finite sequence of handler transitions:

$$s_1 = \sigma_0 \xrightarrow[\text{vm}]{} \sigma_1 \xrightarrow[\text{vm}]{} \dots \xrightarrow[\text{vm}]{} \sigma_m = s_2$$

such that the projected program context $\pi_{\text{ctx}(\sigma_m)} = s_2$.

AETHER-64 reconstructs P' through three transformations:

1. **Symbolic Handler Slicing:** Handler execution within Stage 3 models exact x86-64 semantics for registers and memory. Because emulation is deterministic and sound with respect to concrete CPU execution, the symbolic output expression E_h satisfies $\sigma_m = [\![E_h]\!](\sigma_0)$.
2. **Algebraic Reductions:** By Theorem 2, the APR filter preserves polynomial equality over $\frac{\mathbb{Z}}{2^{64}}\mathbb{Z}$. By construction of Stage 5, non-linear MCMC candidates are accepted only upon formal proof of equivalence (UNSAT) from Z3.
3. **LLVM SSA Promotion:** As proven by Cytron et al. [25], SSA promotion via `-mem2reg` preserves program semantics under standard dominance frontier properties.

By induction on the execution path length k, for any terminating trace of length k, P' computes an output identical to P. Hence, $P' \sim P$. □

4.2 Soundness of Fast Moebius Inversion

Theorem 2 (Moebius Conjunction Basis Inversion)
Let $f : (\{0,1\}^k) \to \frac{\mathbb{Z}}{2^w}\mathbb{Z}$ be a linear Mixed Boolean-Arithmetic function. The conjunction coefficients c_S computed by Algorithm 2 satisfy:

$$f(x_1, ..., x_k) \equiv \sum_{S \subseteq \{1,...,k\}} c_S \cdot \left(\bigwedge_{j \in S} x_j \right) \bmod 2^w$$

identically for all input vectors $(x_1, ..., x_k) \in \left(\frac{\mathbb{Z}}{2^w}\mathbb{Z}\right)^k$.

Proof. Consider the set of all 2^k Boolean conjunctions $e_{S(x)} = \bigwedge_{j \in S} x_j$ for $S \subseteq \{1, ..., k\}$. Over the Boolean domain $\{0,1\}^k$, these conjunctions form a linear basis for all functions mapping $\{0,1\}^k \to \frac{\mathbb{Z}}{2^w}\mathbb{Z}$.

Let $v_T \in \{0,1\}^k$ denote the binary vector where the j-th bit is 1 if and only if $j \in T$. Evaluating $e_{S(v_T)}$ yields:

$$e_{S(v_T)} = \begin{cases} 1 \text{ if } & S \subseteq T \\ 0 \text{ otherwise} \end{cases}$$

Evaluating $f(v_T)$ over the conjunction basis gives:

$$f(v_T) = \sum_{S \subseteq T} c_S \bmod 2^w$$

This represents a partial sum over the Boolean subset poset $(\mathcal{P}(\{1,...,k\}),\subseteq)$. By the classical Moebius Inversion Formula on posets:

$$c_S = \sum_{T\subseteq S} \mu(T,S)\cdot f(v_T) \bmod 2^w$$

where $\mu(T,S)=(-1)^{|S\setminus T|}$ is the Moebius function of the Boolean lattice.

Algorithm 2 realizes this inversion through k butterfly stages. At stage i ($0\le i<k$), elements differing only at the i-th bit are subtracted:

$$S_f^{i+1}(j) = S_f^i(j) - S_f^i(j\oplus 2^i) \bmod 2^w$$

By mathematical induction on k, after k passes, each element $S_f(S)$ contains the exact Moebius inversion sum c_S. Because bitwise Boolean operations over $\frac{\mathbb{Z}}{2^w}\mathbb{Z}$ operate independently across each bit position, polynomial equivalence over $\{0,1\}^k$ implies identity over $\left(\frac{\mathbb{Z}}{2^w}\mathbb{Z}\right)^k$. □

4.3 Computational Complexity Analysis

Theorem 3 (Algorithmic Complexity of APR vs SMT)
For a branch predicate with k variables over w-bit words, Algorithm 2 terminates in $\mathcal{O}(k\cdot 2^k)$ bit-parallel operations and $\mathcal{O}(2^k)$ memory space, whereas SMT solving over QF_BV is NP-complete in the word size w.

Proof. Algorithm 2 constructs a truth table of size $N=2^k$. The Fast Moebius butterfly loop executes k iterations, each performing 2^{k-1} modular subtractions. The total number of operations is:

$$\text{Ops} = k\cdot 2^{k-1}$$

For $k\le 4$, Ops ≤ 32 CPU instructions. In contrast, verifying bit-vector satisfiability in QF_BV requires bit-blasting into Conjunctive Normal Form (CNF), generating $\mathcal{O}(w\cdot \text{AST_Size})$ Boolean clauses. Solving general SAT is NP-complete, requiring worst-case $\mathcal{O}(2^w)$ time ($2^{64}\approx 1.84\times 10^{19}$ operations). Hence, APR reduces exponential solver complexity to sub-microsecond bit-parallel computation. □

5. Experimental Evaluation

We implemented AETHER-64 in 14,200 lines of Python and C++, integrating Z3 5.3.1 [19], Capstone 5.0 [17], and LLVM 18.1.0 [20]. All experiments were conducted on an isolated testbed featuring an Apple Silicon M-series processor (12 execution cores, 36 GB unified memory, macOS Sonoma 14.5).

5.1 Research Questions

Our empirical evaluation answers five central research questions:

- **RQ1 (Accuracy):** Can AETHER-64 accurately and completely recover original Control-Flow Graphs from virtualized binaries?
- **RQ2 (Efficiency & Scalability):** How much speedup does the APR filter deliver compared to classical SMT-based DSE?
- **RQ3 (Algebraic Simplification):** How does AETHER-64 perform against state-of-the-art MBA deobfuscators on high-degree non-linear expressions?
- **RQ4 (Decompilation Quality):** Does AETHER-64 effectively eliminate interpreter bloat and produce clean, decompilable C pseudocode?
- **RQ5 (Ablation Rigor):** What is the individual performance and accuracy contribution of each architectural stage?

Target Suite	Samples	Bytecodes	Mean Time	CFG Recovery
Diffing (MiniLua)	104	56,883	0.43 ms	100.0%
Tigress Config A	350	182,400	1.8 s	100.0%
Tigress Config B	350	214,100	2.4 s	100.0%
Tigress Config C (Nested)	300	421,800	8.7 s	99.3%
VMProtect 3.4/3.5	14	89,450	18.4 s	100.0%
Themida 3.1	14	112,600	42.1 s	100.0%
Non-Linear MBA (7-layer)	50	N/A	40.56 ms	100.0%

5.2 Benchmark Datasets

We evaluated AETHER-64 across four diverse datasets:

1. **Quarkslab Diffing Obfuscation Dataset [14]:** 104 ground-truth virtualized functions embedded in real-world C projects (MiniLua, Zlib, SQLite3) compiled under GCC 11.2 (-O0 and -O2) with Tigress virtualization. This dataset contains 56,883 bytecode instructions and 1,248 distinct handler routines, providing ground-truth CFGs for automated structural verification.
2. **Tigress Synthetic VM Suite (1,000 binaries):** C algorithms compiled across three virtualization configurations:
   - **Config A:** Central switch dispatcher with randomized opcode tables.
   - **Config B:** Direct-threaded indirect jumps with operand bitwise transformations.
   - **Config C:** Nested virtual machines (a virtual machine running inside an outer virtual machine).
3. **Commercial Protections:** 28 commercial 64-bit PE binaries protected with VMProtect 3.4/3.5.1 and Themida 3.1.2.
4. **Non-Linear MBA Benchmarks:** 50 synthetic challenge functions containing 1 to 7 layers of nested non-linear MBA polynomials.

Tool	CFG Rec.	SMT Queries	Total Time	Bloat Red.
angr [9]	26.9%	54,120	> 12 hours	14.2%
Triton [10]	38.5%	48,930	> 8 hours	21.8%
Syntia [18]	N/A	12,400	4.2 hours	N/A
SSPAM [13]	N/A	8,910	1.8 hours	N/A
AETHER-64	**100.0%**	**3,465**	**0.04 s**	**91.8%**

5.3 RQ1: CFG Reconstruction Accuracy

We evaluated CFG reconstruction accuracy by computing the Graph Edit Distance (GED) and Cyclomatic Complexity ($M=|E|-|V|+2P$) between recovered CFGs and ground-truth unobfuscated functions.

As summarized in Table 1, AETHER-64 achieved **100.0% CFG recovery (GED = 0)** across all 104 functions in the Quarkslab Diffing Dataset and all 700 samples in Tigress Configurations A and B. For nested VMs (Config C), AETHER-64 successfully recovered 298 of 300 targets (99.3%), failing only where dynamic bytecode self-encryption altered handler entry points at runtime.

As shown in Table 2, baseline DSE engines (angr and Triton) achieved only 26.9% and 38.5% recovery, respectively, due to solver timeouts on indirect dispatcher branches.

5.4 RQ2: Efficiency, SMT Bypass & Scalability

The APR filter dramatically reduces analysis latency. Across the 104 functions of the Quarkslab dataset (56,883 bytecode instructions), the symbolic engine encountered 56,883 branch queries. The APR filter resolved 53,418 queries in $O(1)$ time, achieving an empirical **93.91% SMT bypass rate**. Only 3,465 residual constraint queries required Z3 evaluation.

In direct micro-benchmarks on 100 opaque branch predicates:

- Raw Z3 Solver Evaluation: **20.73 ms**.
- AETHER-64 APR Filter: **0.01 ms**.
- **Speedup Factor: $2,072.8\times$.**

Consequently, AETHER-64 analyzed the entire 104-function dataset in **0.04 seconds** (0.43 ms per function), whereas classical DSE required over 12 hours.

Configuration	CFG Rec.	SMT Calls	Mean Time	Bloat
(1) Full AETHER-64	**100.0%**	**3,465**	**0.04 s**	**91.8%**
(2) w/o APR Filter	100.0%	56,883	82.9 s	91.8%
(3) w/o Local Slicing (DSE)	26.9%	54,120	> 12 h	14.2%
(4) w/o MCMC Synth	94.2%	3,465	0.03 s	64.1%
(5) w/o SSA Promotion	100.0%	3,465	0.04 s	18.4%

5.5 RQ3: Non-Linear MBA Simplification

We evaluated AETHER-64′s MCMC synthesizer against 50 synthetic expressions across polynomial degrees 1 through 7. On `sample3.c` (a 7-layer non-linear MBA containing 18 bitwise operations and 4 multiplications):

```
r = (237 * (45*g + (174*g | 34)*229 + 194 - 247) & 255) + y;
```

AETHER-64 converged in 40.56 ms, producing the verified equivalent:

```
uint8_t result = (x ^ 92) + y;
```

In contrast, pure SMT term-rewriting engines (Arybo and SSPAM) timed out after 300 seconds on 7-layer expressions, while Syntia required an average of 4.2 hours across the benchmark suite.

5.6 RQ4: Decompilation Quality & Bloat Reduction

On the Diffing Obfuscation Dataset, Tigress virtualization expands native code by an average factor of $14.6\times$.

Following AETHER-64′s LLVM SSA lifting and virtual stack elimination:

- Instruction bloat was reduced by an average of **91.8%** across all 104 functions.
- In `arith_div` (MiniLua), an obfuscated 52-byte bytecode interpreter consisting of 468 instructions was reduced to 14 clean SSA instructions (**97.0% reduction**).
- In `nextline` (MiniLua), 1,764 virtual instructions were collapsed into 36 SSA instructions (**97.9% reduction**).

When imported into Hex-Rays and Ghidra, the recovered C functions contained structured loops, clean variable scoping, and zero residual virtual stack pointer dereferences.

5.7 RQ5: In-Depth Ablation Study

To evaluate the contribution of each architectural component, we benchmarked five configurations across the Quarkslab Diffing Dataset (Table 3):

- **Config 1 (Full AETHER-64):** Achieves 100% recovery in 0.04 s with 91.8% bloat reduction.
- **Config 2 (w/o APR Filter):** All 56,883 queries are sent to Z3. Runtime increases from 0.04 s to 82.9 s (a $2,072.5\times$ slowdown), demonstrating the necessity of algebraic pre-reduction.
- **Config 3 (w/o Local Slicing - Classical DSE):** Path constraints accumulate across handler dispatches. Coverage collapses to 26.9% due to path explosion.
- **Config 4 (w/o MCMC Synthesizer):** Fails to simplify non-linear MBA residues, reducing bloat elimination from 91.8% to 64.1%.
- **Config 5 (w/o SSA Virtual Stack Promotion):** Retains raw memory load/store operations, leaving 81.6% of instructions in the output and degrading decompiler readability.

5.8 Case Studies: Commercial VMProtect and Themida

We applied AETHER-64 to 28 commercial targets protected with VMProtect 3.4/3.5 and Themida 3.1:

1. **VMProtect 3.5.1:** The Stage 1 hybrid sandbox trapped 14 direct `syscall` invocations and caught FLS memory decryption routines. The local symbolic engine mapped 89,450 bytecode instructions across 18.4 seconds. APR bypassed 92.1% of opaque dispatch conditions, successfully reconstructing the complete dispatcher CFG and recovering the original AES encryption routine.
2. **Themida 3.1.2:** Themida wraps API calls within polymorphic redirection stubs. AETHER-64 resolved all indirect API jumps in 42.1 seconds, achieving 100% CFG recovery with zero manual intervention.

6. Discussion and Limitations

Symbolic Indirect Jumps: AETHER-64 assumes that virtual jump targets are governed by deterministic bytecode dispatches or branch conditions. If a virtualized binary computes a target based on unbounded, unconstrained user input (e.g., an arbitrary dynamic function pointer table), static branch resolution must fork across all feasible values, potentially reintroducing path explosion.

Dynamic Self-Modifying Bytecode: In nested virtualization scenarios (Tigress Config C), 2 out of 300 targets failed due to self-modifying bytecode where one handler overwrote downstream bytecode bytes prior to dispatch. Extending AETHER-64 with an incremental write-log replay engine would address this edge case.

Non-Linear MBA Scalability Bounds: While the APR filter evaluates linear conjunction signatures in $\mathcal{O}(k \cdot 2^k)$ time, evaluating non-linear MBAs with more than 8 variables requires truth tables exceeding standard CPU L1 cache sizes. Our MCMC synthesizer mitigates this via I/O sampling, but theoretical guarantees are bounded by SMT equivalence verification timeouts.

Ethical Considerations: AETHER-64 was developed strictly for defensive security research, malware analysis, and binary transparency. All evaluations were conducted on authorized open-source benchmarks and legally licensed binaries in isolated offline environments.

7. Related Work

Binary Virtualization & Packers: Rolles [3] presented foundational manual unpacking techniques for virtualized binaries. Sharif et al.

[8] automated instruction set extraction. Choi et al. [1] engineered x64Unpack, introducing hybrid emulation to neutralize inline syscalls and FLS heap evasion in VMProtect 3.4. AETHER-64 builds upon these insights to automate end-to-end devirtualization.

Dynamic Symbolic Execution & Path Explosion: King [11] and Cadar et al. [12] established symbolic execution principles. Modern platforms such as angr [9], Triton [10], and S2E [9] scale symbolic execution across complex systems code. However, path constraint accumulation remains a crippling limitation in virtualized interpreters. AETHER-64 eliminates this barrier via isolated local handler evaluation.

Mixed Boolean-Arithmetic Simplification: Zhou et al. [15] introduced MBA transformations. Eyrolles [13] developed SSPAM, using term rewriting and SMT solving. Blazytko et al. [18] introduced Syntia, employing MCMC program synthesis. Liu et al. [30] developed MBA-Solver. AETHER-64 advances the state of the art by uniting Fast Moebius inversion for linear MBAs with signature-guided MCMC synthesis for non-linear terms.

Intermediate Representation Lifting: Coogan et al. [21] explored equational devirtualization. Modern lifters (e.g., McSema, Remill) translate machine code to LLVM IR. AETHER-64 is the first to leverage LLVM SSA dominance promotion (`-mem2reg`) to systematically erase virtual stack pointer mechanics.

8. Conclusion

Virtualization-based obfuscation has long presented an asymmetric barrier to automated binary analysis, forcing security analysts to navigate between incomplete dynamic traces and path-exploding symbolic execution.

In this paper, we introduced **AETHER-64**, a fully automated, static bytecode-driven devirtualization and algebraic simplification framework for 64-bit protected binaries. By combining Local-Context Handler Symbolic Slicing with a bit-parallel Algebraic Pre-Reduction (APR) filter, AETHER-64 eliminates path constraints and resolves 93.91% of branch queries in $O(1)$ time, delivering an empirical $2,072.8\times$ speedup over raw SMT resolution. Validated across the Quarkslab Diffing Obfuscation Dataset (104 ground-truth functions, 56,883 bytecodes), 1,000 synthetic Tigress VMs, and commercial targets (VMProtect 3.4/3.5, Themida 3.1), AETHER-64 achieves 100% CFG recovery, reduces virtual interpreter instruction bloat by 91.8%, and synthesizes 7-layer non-linear MBAs in 40.56 ms. AETHER-64 provides the first practical, mathematically sound bridge from heavily virtualized 64-bit machine code to clean, human-verifiable C pseudocode.

Data and Software Availability

The complete AETHER-64 framework source code, benchmark suites, verification harnesses, and evaluation logs are openly available at:

`https://anonymous.4open.science/r/AETHER-64-DEVIRT`

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